



VECTOR SPACES

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CLASS 1 : CONTENT



- Definition of Vector Space
- Examples of Vector Space
- Definition of Subspace
- Examples of Subspaces

VECTOR SPACES : DEFINITION



A **Real vector space** V is a nonempty set of objects called **vectors**, together with (**Scalar multiplication and Vector addition**) satisfying the following axioms :

- I. If $u, v \in V$, then $u + v \in V \Rightarrow V$ is closed under vector addition.
- II. If $c \in \mathbb{R}$ & $u \in V$, then $cu \in V \Rightarrow V$ is closed under scalar multiplication.

These operations satisfy the following properties for $u, v, w \in V$ & c_1, c_1 are scalars

- a) $u + v = v + u$ (commutative law)
- b) $u + (v + w) = (u + v) + w$ (Associative law)

VECTOR SPACES

c) there is a unique zero vector i.e., 0 such that $0 + u = u + 0 = u$

(identity law) Additive identity ' 0 ' $\in V$

d) for each u there is a unique vector $(-u)$ such that

$u + (-u) = (-u) + u = 0$ (Inverse law)

e) $c_1(u + v) = c_1u + c_1v$

f) $(c_1 + c_2)u = c_1u + c_2u$

g) $(c_1 + c_2)u = c_1u + c_2u$

h) $1u = u$, Where 1 is a multiplicative identity s.t. $1 \in \mathbb{R}$

VECTOR SPACES: EXAMPLES



Example 1 The following are examples of vector spaces:

1. The set of all real number $\mathbb{R}$ associated with the addition and scalar multiplication of real numbers.
2. The set of all the [complex numbers](#) $\mathbb{C}$ associated with the addition and scalar multiplication of complex numbers.
3. The set of all [polynomials](#) $R_n(x)$ with real coefficients associated with the addition

VECTOR SPACES: EXAMPLES



4. The set of all vectors of dimension n written as $\mathbb{R}^n$ associated with the addition and scalar multiplication as defined for 3-d and 2-d vectors for example.
5. The set of all [matrices](#) of dimension $m \times n$ associated with the addition and scalar multiplication as defined for matrices.

VECTOR SPACES: EXAMPLES

Example 1 :

Prove that the set of all 2 by 2 matrices associated with the matrix addition and the scalar multiplication of matrices is a vector space.

Solution: Consider $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $A' = \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix}$ s.t. $A, A' \in V$
and $r, s \in R$

Let V be the set of all 2 by 2 matrices.

1) Addition of matrices gives

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} = \begin{bmatrix} a + a' & b + b' \\ c + c' & d + d' \end{bmatrix}$$

Adding any 2 by 2 matrices gives a 2 by 2 matrix and therefore the result of the addition

VECTOR SPACES: EXAMPLES

Scalar multiplication of matrices gives

$$r \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ra & rb \\ rc & rd \end{bmatrix}$$

Multiply any 2 by 2 matrix by a scalar and the result is a 2 by 2 matrix is an element of V .

3) Commutativity

$$\begin{aligned} & \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} \\ &= \begin{bmatrix} a + a' & b + b' \\ c + c' & d + d' \end{bmatrix} \\ &= \begin{bmatrix} a' + a & b' + b \\ c' + c & d' + d \end{bmatrix} \\ &= \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} + \begin{bmatrix} a & b \\ c & d \end{bmatrix} \end{aligned}$$

4) Associativity of vector addition

$$\begin{aligned} & \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} \right) + \begin{bmatrix} a'' & b'' \\ c'' & d'' \end{bmatrix} \\ &= \begin{bmatrix} a + a' & b + b' \\ c + c' & d + d' \end{bmatrix} + \begin{bmatrix} a'' & b'' \\ c'' & d'' \end{bmatrix} \\ &= \begin{bmatrix} (a + a') + a'' & (b + b') + b'' \\ (c + c') + c'' & (d + d') + d'' \end{bmatrix} \\ &= \begin{bmatrix} a + (a' + a'') & b + (b' + b'') \\ c + (c' + c'') & d + (d' + d'') \end{bmatrix} \\ &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \left(\begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} + \begin{bmatrix} a'' & b'' \\ c'' & d'' \end{bmatrix} \right) \end{aligned}$$

VECTOR SPACES: EXAMPLES

5) Associativity of multiplication

$$\begin{aligned} r \left(s \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) &= r \left(\begin{bmatrix} sa & sb \\ sc & sd \end{bmatrix} \right) \\ &= \begin{bmatrix} rsa & rsb \\ rsc & rsd \end{bmatrix} \\ &= \begin{bmatrix} (rs)a & (rs)b \\ (rs)c & (rs)d \end{bmatrix} \\ &= (rs) \begin{bmatrix} a & b \\ c & d \end{bmatrix} \end{aligned}$$

6) Zero vector

$$\begin{aligned} \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \\ = \begin{bmatrix} a+0 & b+0 \\ c+0 & d+0 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \end{aligned}$$

7) Negative vector

$$\begin{aligned} \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} -a & -b \\ -c & -d \end{bmatrix} \\ = \begin{bmatrix} a+(-a) & b+(-b) \\ c+(-c) & d+(-d) \end{bmatrix} \\ = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

VECTOR SPACES: EXAMPLES

8) Distributivity of sums of matrices:

$$\begin{aligned} & r \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} \right) \\ &= \begin{bmatrix} r(a + a') & r(b + b') \\ r(c + c') & r(d + d') \end{bmatrix} \\ &= \begin{bmatrix} ra + ra' & rb + rb' \\ rc + rc' & rd + rd' \end{bmatrix} \\ &= r \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) + r \left(\begin{bmatrix} a' & b' \\ c' & d' \end{bmatrix} \right) \end{aligned}$$

9) Distributivity of sums of real numbers:

$$\begin{aligned} (r + s) \begin{bmatrix} a & b \\ c & d \end{bmatrix} &= \begin{bmatrix} (r + s)a & (r + s)b \\ (r + s)c & (r + s)d \end{bmatrix} \\ &= \begin{bmatrix} ra + sa & rb + sb \\ rc + sc & rd + sd \end{bmatrix} \\ &= \begin{bmatrix} ra & rb \\ rc & rd \end{bmatrix} + \begin{bmatrix} sa & sb \\ sc & sd \end{bmatrix} \\ &= r \begin{bmatrix} a & b \\ c & d \end{bmatrix} + s \begin{bmatrix} a & b \\ c & d \end{bmatrix} \end{aligned}$$

10) Multiplication by 1.

$$1 \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1a & 1b \\ 1c & 1d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

VECTOR SPACES: EXAMPLES



Example 2 :

Show that the set of all real polynomials with a degree $n \leq 3$ associated with the addition of polynomials and the multiplication of polynomials by a scalar form a vector space.

Solution

The addition of two polynomials of degree less than or equal to 3 is a polynomial of degree less than or equal to 3.

The multiplication, of a polynomial of degree less than or equal to 3, by a real number results in a polynomial of degree less than or equal to 3

VECTOR SPACES: EXAMPLES



Hence the set of polynomials of degree less than or equal to 3 is closed under addition

and scalar multiplication (the first two conditions above).

The remaining 8 rules are automatically satisfied since the polynomials are real.

VECTOR SPACES: EXAMPLES



Example 3 :

Show that the set of integers associated with addition and multiplication by a real number

IS NOT a vector space

Solution :

The multiplication of an integer by a real number may not be an integer.

Example: Let $x = -2$

If you multiply x by the real number $\sqrt{3}$ the result is NOT an integer.

VECTOR SPACE

Few examples :

1. $\mathbb{R}$ = the set of all real numbers

$$2. \mathbb{R}^2 = \{ (x, y) / x, y \in \mathbb{R} \}$$

$$3. \mathbb{R}^3 = \{ (x, y, z) / x, y, z \in \mathbb{R} \}$$

$$4. \mathbb{R}^n = \{ (x_1, x_2, \dots, x_n) / x_i \in \mathbb{R} \}$$

$$5. \mathbb{R}^\infty = \{ (x_1, x_2, \dots) / x_i \in \mathbb{R} \}$$

VECTOR SPACES: EXAMPLES



Problem 1 :

Verify whether the following

$$V = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} ; x \geq 0, y \geq 0, x, y \in \mathbb{R} \right\}$$

Is a vector space

under usual vector addition and scalar multiplication.

VECTOR SPACES

- Closure property holds good.
- Associative property holds
- $\exists 0 \in V \exists u + 0 = u = 0 + u$,
- $\forall u \in V \exists -u \notin V$ $\left[\text{Eg : } u = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \in V, -u = \begin{bmatrix} -1 \\ -2 \end{bmatrix} \notin V \right]$

Therefore Inverse law doesn't hold

Hence V is not a vector space.

Problem 2 : $V = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 / x + y = 0 \right\} u, v \in V$

All the properties holds good

Hence V is a vector space

VECTOR SPACES



Precisely ,

We can add any two vectors and we can multiply all vectors by scalars. In other words, we can take linear combinations.

SUBSPACES : DEFINITION



SUBSPACES

A nonempty subset of a vector space is called a subspace of V , if it is itself a vector space under the same operations of vector addition and scalar multiplication as defined in vector space.

The following are the properties satisfied by a subspace of V

- i) $0 \in W$ (zero vector always belongs to a subspace)
- ii) if $u, v \in W$ Then $u + v \in W$
- iii) If ' c ' is a scalar and $u \in W$ then $cu \in W$

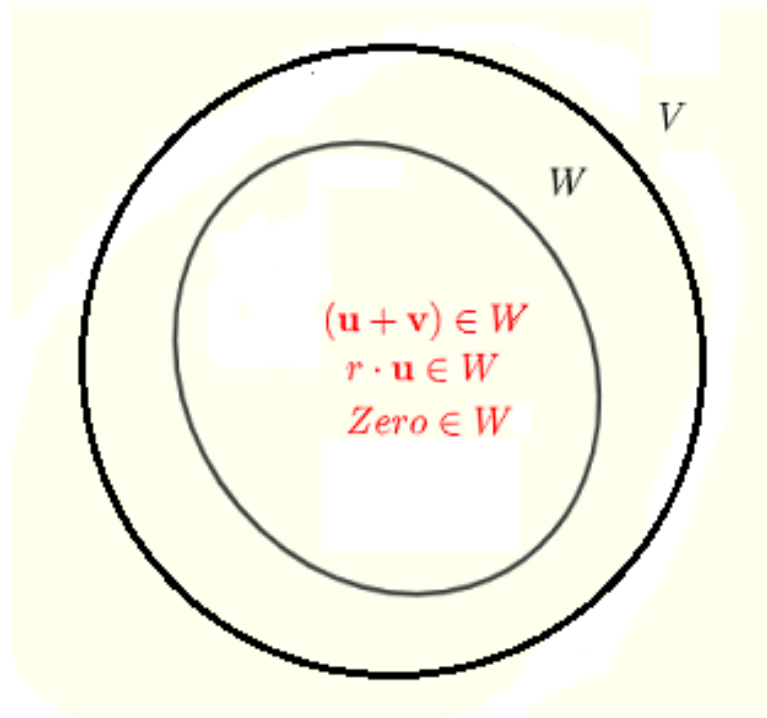
SUBSPACES : DEFINITION

If W is a subset of a [vector space](#) V and if W is itself a vector space under the inherited operations of addition and scalar multiplication from V , then W is called a subspace

To show that the W is a subspace of V , it is enough to show that

1. W is a subset of V
2. The zero vector of V is in W
3. For any vectors $\mathbf{u}$ and $\mathbf{v}$ in W , $\mathbf{u} + \mathbf{v}$ is in W . (closure under addition)
4. For any vector $\mathbf{u}$ and scalar r , $r \cdot \mathbf{u}$ is in W . (closure under scalar multiplication).

SUBSPACES : DEFINITION



SUBSPACES : EXAMPLES

Example 1

The set W of vectors of the form $(x, 0)$ where $x \in \mathbb{R}$ is a subspace of $\mathbb{R}^2$ because:

W is a subset of $\mathbb{R}^2$ whose vectors are of the form (x, y) where $x \in \mathbb{R}$ and $y \in \mathbb{R}$

The zero vector $(0, 0)$ is in W

$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0)$, closure under addition

$r \cdot (x, 0) = (rx, 0)$, closure under scalar multiplication

SUBSPACES : EXAMPLES



Example 2

The set W of vectors of the form (x, y) such that $x \geq 0$ and $y \geq 0$ is not a subspace of

$\mathbb{R}^2$ because it is not closed under scalar multiplication.

Vector $\mathbf{u} = (2, 2)$ is in W but its negative $-1(2, 2) = (-2, -2)$ is not in W .

SUBSPACES :

- Note : If U and W are two subspaces of a vector space V , intersection $U \cap W$ is also a subspace of V .
 $0 \in U$ and $0 \in W$ since U and W are subspaces they must contain '0' . $0 \in U \cap W$
- The intersection of any number of subspaces of a vector space V is a subspace of V

SUBSPACES

Subspace of $\mathbb{R}^3$

- i. $\mathbb{R}^3$ itself
- ii. zero vector i.e., $\left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$
- iii. line passing through origin
- iv. plane passing through origin
- v. In general, if $V = \mathbb{R}^n$, the possible subspaces are , lines through origin, 2-d planes through origin, 3-d planes through origin, , (n-1)- d planes through origin and the space itself .



THANK YOU

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